

freMTPL2 French Motor Third-Part Liability dataset

Description

In the two datasets, **freMTPL2freq** and **freMTPL2sev**, risk features are collected for 677,991 motor third-party liability policies (observed mostly on one year).

- **freMTPL2freq** contains the risk features and the claim number.
- **freMTPL2sev** contains the claim amount and the corresponding policy ID.
- Some claim amounts are fixed claim amounts based on the French IRSA-IDA claim convention.

Usage

```
data(freMTPL2freq)
```

```
data(freMTPL2sev)
```

freMTPL2freq (11 columns)

Column Name	Description
IDpol	The policy ID (used to link with the claims dataset).
ClaimNb	Number of claims during the exposure period.
Exposure	The period of exposure for a policy, in years.
VehPower	The power of the car (ordered values).
VehAge	The vehicle age, in years.
DrivAge	The driver age, in years (in France, people can drive a car at 18).
BonusMalus	Bonus/malus, between 50 and 350: <100 means bonus, >100 means malus in France.
VehBrand	The car brand (unknown categories).
VehGas	The car gas, Diesel or regular.

Area	The density value of the city community where the car driver lives in: from "A" for rural area to "F" for urban centre.
Density	The density of inhabitants (number of inhabitants per square-kilometer) of the city where the car driver lives in.
Region	The policy region in France (based on the 1970-2015 classification).

freMTPL2sev (2 columns)

Column Name	Description
IDpol	The occurrence date (used to link with the contract dataset).
ClaimAmount	The cost of the claim, seen as of a recent date.

Source

Unknown private insurer.

References

- M. Denuit, A. Charpentier, and J. Trufin (2021), Autocalibration and tweedie-dominance for insurance pricing with machine learning, *Insurance: Mathematics and Economics*, 101, 485–497, doi:10.1016/j.insmatheco.2021.09.001.
- T. Miljkovic and D. Fernández (2018). On Two Mixture-Based Clustering Approaches Used in Modeling an Insurance Portfolio, *Risks*, 6(2), 1–18, doi:10.3390/risks6020057.
- A. Noll, R. Salzmänn, and M. V. Wuthrich (2020), Case study: French motor third-party liability claims, *Innovation Practice eJournal*, doi:10.2139/ssrn.3164764.
- N. Pocuca, P. Jevtic, P. D. McNicholas, and T. Miljkovic (2020), Modeling frequency and severity of claims with the zero-inflated generalized cluster-weighted models, *Insurance: Mathematics and Economics*, 94, 79–93, doi:10.1016/j.insmatheco.2020.06.004.

Examples

(1) load of data

```
data(freMTPLfreq)
```

```
dim(freMTPLfreq)
```

```
data(freMTPLsev)
```

```
dim(freMTPLsev)
```

(2) check

should be equal

```
sum(freMTPLsev$PolicyID %in% freMTPLfreq$PolicyID)
```

```
sum(freMTPLfreq$ClaimNb)
```

(1) load of data

```
data(freMTPL2freq)
```

```
dim(freMTPL2freq)
```

```
data(freMTPL2sev)
```

```
dim(freMTPL2sev)
```

<https://cas.uqam.ca/pub/web/CASdatasets-manual.pdf>

<https://dutangc.github.io/CASdatasets/>

<https://www.kaggle.com/datasets/karansarpal/fremtpl2-french-motor-tpl-insurance-claims/data>

<https://www.openml.org/search?type=data&sort=runs&id=43593&status=active>